

Marco Mo

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SUMMARY

Data Engineer delivering Python/SQL ETL, database integrations and production analytics. Combines schema validation, privacy controls and human review to turn diverse sources into validated payloads, dashboards and reporting tools for staff and managers.

EXPERIENCE

Process Optimization, Automation & AI Specialist

Aug 2026 – Present

Multi-Post

- Consolidated twelve workflows for preparation, reporting and audits into a staff-used application.
- Automated spreadsheet and fixed-width preparation with validation, deduplication and review.
- Reduced selected artwork preparation tasks from approximately 30 minutes or more to seconds.
- Delivered production dashboards and machine-event reporting from department-manager requirements.

Junior AI Specialist

Jun 2026 – Aug 2026

ADC Nederland

- Developed four systems for order ingestion, database access, chatbots and process mining.
- Translated workflows and database constraints into solutions with ICT colleagues and directors.
- Combined typed APIs, SQL analytics, permissions, privacy controls and automated testing.

PROJECTS

Omnimap — Multi-Format Order Ingestion and Migration Platform

- Designed ETL converting XML, JSON, CSV and fixed-width orders into validated API payloads.
- Unified canonical schemas, versioned mappings and lookup tables through a shared engine.
- Completed 6,044 validated conversions in 86.6 seconds, detecting 1,104 duplicate files.
- Added Pydantic validation, Alembic migrations, batch reporting and manual exception review.

MCP Server for the Wihabo Order Database

- Built a FastMCP server exposing 67 structured tools over REST and read-only MySQL.
- Enforced tenant-scoped authentication, PII redaction and privacy regression tests.
- Added typed schemas and aggregate SQL tools to avoid loading complete order histories.

Flowcenter Production Process Mining and Operational Analytics

- Built privacy-aware extraction over MySQL production logs using batches and scan events.
- Measured delays, rework, workload and data quality to prioritize production burdens.
- Delivered Parquet outputs and a dashboard with product, component and transition filters.

EDUCATION

MSc Data Science and Artificial Intelligence

2023 – 2025

Eindhoven University of Technology

- Thesis: Reward-Free Safe Reinforcement Learning Exploration Using Different Entropy Measures

BSc Data Science

2020 – 2023

Eindhoven University of Technology and Tilburg University

- End project: Health Platform Text Classification Using Active Learning

SKILLS

Programming: Python, SQL, Bash, C++

Data engineering: ETL, batch processing, schema validation, deduplication, REST APIs, FastAPI, Pydantic, Alembic, Docker Compose, pytest

Databases and formats: PostgreSQL, MySQL, SQLite, XML, JSON, CSV, Parquet, fixed-width files

Analytics and delivery: pandas, process mining, operational analytics, dashboards, Django, GitHub Actions

LANGUAGES

English (Fluent), Cantonese (Native), Mandarin (Fluent), Dutch (A2)